Assignment 5 report

US Midterm Issues, Sentiment, Democratic Sweep Odds and a Mid-Term Basket: Jul 1 to Oct 7, 2026

FRE-GY 7871 A · NLP and the Investment Process

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1. What I did

The midterms are on Nov 3, 2026. Republicans hold the White House, the House (218 R, 214 D, 1 I) and the Senate (53 R, 47 D incl. 2 independents), so a "Democratic sweep" needs a net gain of 3 House seats and 4 Senate seats (VP Vance breaks a 50-50 tie). My window is Jul 1 to the Oct 7 close (99 calendar days, 69 trading days).

Data. From social media I collected **31,540 tweets** (twitterapi.io) and, in addition to Twitter, also **143,625 Reddit posts and comments** from six political subreddits (Arctic Shift); 25,743 and 40,148 survive the filters. From news, **72,688 unique election headlines** (Google News RSS): 69,282 from 30 named outlets and 3,406 from Google News' US feed (1,118 other sites). The sweep odds are Polymarket's "Democrats Sweep" contract and Kalshi's `KXBALANCEPOWERCOMBO-27FEB-DD`, hourly, snapshotted at 16:00 ET. Prices for 27 stocks and 10 ETFs or futures come from yfinance. I ran BERTopic with an LLM (Claude subagents) naming the issues, built a validated directional index (+ = more pro-Democrat or anti-GOP), ran pre-registered tests against the daily change in the odds, and fixed a 17-name basket before seeing any test-window return. To prevent the mistakes and shortcomings of my earlier assignments, the analysis, the results and the prose were all checked before submission: every number against the data, every formula against its source, and independent reviewer agents (Claude subagents) recomputed the results.

Table 1. Event calendar (ET; all checked against news sources)

Date	Event
Feb 28 to Sept 13	Iran war starts (ongoing); Section 301 tariffs replace Section 122 (Jul 24); funding to Dec 11 signed (Sept 2); sweep odds jump 49.5% to 53.5% (Sun Sept 13)
Sept 15 to 23	NYT/Siena poll D 51 to 43, CLARITY Act fails cloture, Fed hikes 25bp (Sept 15 to 16); Trump approval 32% (Reuters/Ipsos), Cook shifts GA, KS, SC toward D (Sept 21 to 23)

Short answers.

- **Q1 (issues):** in online political posters' own posts, foreign policy (mostly Israel and the Iran war) and democracy are top three in every model version, and the economy gets only 5% to 9% of issue talk, far below polls (22% to 31%) and news (14.8%); the finer order moves with the run.
- **Q2 (sentiment module):** a directional index, fair per post (macro-F1 0.60, κ 0.38 on 110 blind-labelled posts vs 0.22 for a guess), mostly noise per day (split-half r 0.08).
- **Q3 (sweep odds):** no. The index and the daily change in P(sweep) correlate at r = 0.067 (N = 99), and none of the 7 pre-registered tests passes (smallest BH q 0.31).
- **Q4 (basket):** the sentiment index and the basket's market-adjusted return are uncorrelated (r = -0.02, N = 69). Its +6.4% window total is an endpoint effect: -0.3% on Oct 1, after the odds' whole 21-point rise, then +6.6% in Oct 2 to 7 while the odds ended where they started (63.5%).
- **Q5:** what posters argue about is measurable and differs from polls and media, but daily political sentiment moves with neither the sweep odds nor exposed stocks.

2. Data

- **Twitter/X** (twitterapi.io): one query on midterms, Congress, the parties, "vote blue/red" and House or Senate races, `min_faves:5` (a probe showed 51% of `min_faves:2` tweets pass the views rule vs 95%); one page (max 20 tweets, a median of 2 minutes of posts) per 90-minute window, i.e. 16 short snapshots a day. **Reddit, in addition to Twitter** (Arctic Shift): r/politics, r/Conservative, r/democrats, r/Republican, r/PoliticalDiscussion and r/moderatepolitics; every post plus the first 50 comments of each 6-hour window.
- **News** (Google News RSS, election query only): `site:` pulls for 30 outlets in Sacerdote-style groups, US left 7 (CNN, MS NOW, NYT...), US right 6 (Fox, NY Post, Breitbart...), US centre 9 (AP, Reuters, Politico...), business 4 and international 4 (full list in the README), plus the US-edition feed. A feed returns at most 100 items, and 352 of 445 capped windows stay capped even by day, so an outlet-day is a relevance-ranked sample. Headlines carry a date only (78% at 07:00 GMT), so news is never used in a timed test.
- As discussed in class, raw social posts are very noisy, so to keep the signal-to-noise ratio under control (after cleaning as in Bagheri and Yener) a post is kept only if it is in the window, not removed, English, has **a US-politics term in the text itself**, no spam or promo, 5+ words, passes an engagement rule (**tweets: 100+ views, 10+ followers, account 30+ days; Reddit: score 2+**), is not a near duplicate (MinHash, Jaccard ≥ 0.8), and is one of at most 5 posts by that author that day. Of 175,165 downloaded items, 65,891 survive; the politics-term rule removes the most (153,306 to 99,413), then the engagement rule (98,192 to 67,559). They split into **voter voice** (comments, self-posts, non-media tweets: 45,278) and **shared media** (Reddit link posts, whose r/politics titles are the exact headline, and news-account tweets: 20,613).
- **Prediction markets.** Polymarket's "Democrats Sweep" resolves on seat counts (independents count with their caucus; a 50-50 Senate goes to the Vice President's party); Kalshi's DD pays if the Speaker and the Senate President pro tempore are Democrats on Feb 1, 2027. Polymarket's price matched its order-book midpoint in a live check; about a quarter of Kalshi's hourly candles have no trade (15% in the window), so I use Kalshi's bid/ask mid (on 2 days the latest candle is an hour old). Both were at 63.5% on Oct 7; the gap stays within -1.0 to +6.0 points (mean +1.7).
- **Equities** (yfinance, official closes): 27 names in 9 policy groups, SPY, and factor ETFs (ITA, XLV, IBIT, USO, ICLN, XRT).

3. Method

- **Topic model.** BERTopic (Grootendorst 2022): sentence embeddings (MiniLM), UMAP (5 dims), HDBSCAN clusters (min cluster 240 = 0.4% of a 60,001-doc sample stratified by day and source) and c-TF-IDF words; fitted on social and news together, applied to all 138,579

docs. The baseline is **LDA** (latent Dirichlet allocation; Blei, Ng and Jordan 2003), the model of the class reading Bybee et al. (2024): each document is a mix of topics and each topic a distribution over words, learned from word counts only. On a 30k subsample BERTopic is more coherent at every seed (C_NPMI 0.065 to 0.071, C_V 0.52 to 0.53) than LDA at its best K = 10 (C_NPMI 0.0001, C_V 0.48); over 3 seeds the doc assignments agree at ARI 0.59 to 0.63.

- **The LLM step.** A Claude subagent with no context saw only each topic's top words and 5 example docs and mapped the 61 topics to 14 of the 16 allowed issues (none to political violence or energy) or to horse race, media or noise (34 issue topics, 27 non-issue; 37 high, 19 medium, 5 low confidence). India's Congress-party topic (3,803 docs, 89% of them tweets mentioning Congress) went to noise. Against separate blind issue labels on the 150 validation items, topic plus taxonomy matches on 72% of the 85 items the model assigns.
- **Direction tools.** Plain tone does not say which party gains, so each tool gives a direction score s in [-1, 1]: (1) **zero-shot NLI** (natural language inference: the model judges whether a statement such as "This text supports the Democrats or criticizes Republicans or Trump" follows from the post; Laurer et al. 2023), s = P(pro-D statement) - P(pro-R statement); (2) **target-signed Twitter-RoBERTa**, s = (P_pos - P_neg) x τ, τ = +1 if only Democrat-side names appear, -1 if only GOP-side names (Trump, MAGA, Vance...); (3) target-signed VADER; (4) an LLM-distilled logistic regression on the embeddings, trained on 1,500 subagent labels; (5) plain RoBERTa tone, the non-directional baseline.
- **Validation.** 150 items (60 tweets, 50 Reddit, 40 headlines, 2/3 of them naming a party side) were labelled blind by a fresh Claude subagent that saw only the guide and the text. **These are LLM labels, not human labels.** A second subagent relabelled 50 calibration items: κ = 0.96 (98% agreement).

Table 2. Direction validation, macro-F1 [95% bootstrap CI] (κ)

Text	NLI	RoBERTa x τ	VADER x τ	LLM-distilled
Tweets (60)	**0.59** [0.46, 0.71] (0.40)	0.54 (0.35)	0.48 (0.25)	0.55 (0.35)
Reddit (50)	0.47 (0.16)	**0.58** [0.39, 0.73] (0.32)	0.47 (0.21)	0.39 (0.10)
Headlines (40)	0.29 (0.03)	0.47 (0.38)	0.43 (0.28)	0.48 (0.38)
All (150)	0.54 (0.29)	0.59 [0.49, 0.67] (0.38)	0.48 (0.26)	0.53 (0.33)

The index's combination (NLI on tweets, RoBERTa x τ on Reddit) scores **macro-F1 0.60 [0.50, 0.70], κ 0.38** on the 110 social items vs 0.22 for always guessing "neither" (accuracy 0.63 vs 0.51; Spearman with the labels 0.43): fair. *Why this result:* NLI reads a whole sentence's stance; target-signing gives 0 to the 60% of Reddit posts naming neither or both parties; the distilled tool (0.63 in its own cross-validation) learned the labeller; tone (0.71) is easier than direction.

Table 3. Which tool is used where, and why

Choice	Used for	Why (evidence)
BERTopic	topics in all text	built for short texts; more coherent than LDA at every K; stable over seeds
LDA	baseline only	the class reading's model, made for long articles; a 20-word post has too few co-occurring words
Claude subagents	topic names; labels	needs reading; blind to scores; non-human, disclosed
NLI	direction of tweets	best on tweets (0.59); handles a tweet that attacks one side without naming the other
RoBERTa x τ	direction of Reddit posts	best on Reddit (0.58 vs NLI 0.47); its tone part is reliable (0.71)
VADER, distilled, tone	robustness only	lower F1 (0.48, 0.53); tone has no direction
Headlines	Q1 shares, E1	dated but not timed, so not usable against 16:00 prices
Polymarket primary, **Kalshi** check	P(sweep)	history from Jul 19, 2025 (Kalshi: Dec 11) covers the beta pre-window; moves on more days (43% vs 39%); resolves on seat counts; Kalshi reruns family A

- **The index** (pre-registered at 20:45 ET on Oct 7, construction details added at 21:28 ET, both before any index existed). Voter-voice docs only. For issue k, source g (Twitter and each subreddit) and day t: the mean s, demeaned within issue and source over the window (so a sub's permanent lean adds nothing), averaged over sources with fixed doc-share weights: I_k,t. The composite is C_t = Σ w_k I_k,t / Σ w_k, w_k the Q1 voter shares of the 8 key issues. Expected sign: **C_t up goes with the sweep odds up**. *Deviation, disclosed:* the pre-registered rule needed 20 docs per issue-day on 90% of days and under the main topics no issue reached it (best 69%) and under outlier-reduced topics two did, too few for 5 to 8 key topics, so index docs use the outlier-reduced topic assignment and a minimum of 5 docs (decided at 21:50 ET on doc counts only).
- **Timing.** Day t is [16:00 ET on t-1, 16:00 ET on t); the odds are the last hourly point at or before 16:05 ET (Polymarket stamps 16:00 points about 16 s late); every doc in day t is asserted to precede that day's price; for stocks, weekends pool into Monday.
- **Tests.** Family A (pre-registered): the same-day regression of ΔP_t on C_t plus Granger tests at lags 1 to 3 both ways (7 tests), BH at q = 0.10 on Newey-West HAC p-values (no small-sample correction), with day-of-week dummies. A claim needs the expected sign, $q < 0.10$, a circular-shift permutation $p < 0.10$, and survival without any one day and without Sept 10 to 26. A robustness grid repeats this for Kalshi, the venue average, Δlogit, the trading-day calendar, 3- and 5-day changes and eight index versions (incl. the pre-registered 20-doc minimum and one adding the economy); its hit count is judged against a joint null that circularly shifts every index series.

Table 4. Formula register (each checked against its source)

Measure	Formula	Source
Issue share; agenda distance	docs in issue k / docs in any issue; TVD = ½ Σ_k abs(a_k - b_k)	Bybee et al. (2024); Levin, Peres and Wilmer (2009) §4.1
Agreement; reliability	κ = (p_o - p_e)/(1 - p_e), macro-F1 = mean per-class F1; split-half 2r/(1+r)	Cohen (1960), scikit-learn; Spearman (1910), Brown (1910)
Sweep change; detectable r	ΔP_t = 100 (p_t - p_t-1); r* = tanh((z_.975 + z_.80)/√(N-3))	Knight (2006), Snowberg et al. (2007); Hulley et al. (2013)
Granger; HAC; FDR	F = ((SSR_r - SSR_u)/q)/(SSR_u/df); Bartlett 1 - j/(m+1); largest k with p_(k) ≤ (k/m) q	Granger (1969); Newey and West (1987); Benjamini and Hochberg (1995)
Abnormal return; election beta	AR = R - α - βR_m (pre-window; test window α = 0); AR_t = a + γ ΔP_t + e	MacKinlay (1997) eq. 3, 7; Knight (2006) eq. 3

Measure	Formula	Source
Rigobon-Sack estimate	d = [Cov_H(ΔP,y) - Cov_L(ΔP,y)]/[Var_H(ΔP) - Var_L(ΔP)]	Rigobon and Sack (2003) eq. 10

4. Q1: Which issues do voters care most about?

Readings used: Bybee et al. (2024) for topic shares as "attention"; Sacerdote, Sehgal and Cook (2020) for comparing outlet groups.

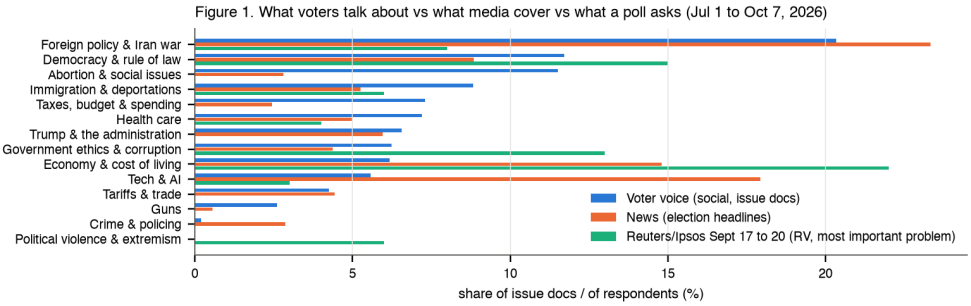


Figure 1. Share of issue docs per issue (horse race, media, noise, outliers excluded), Jul 1 to Oct 7, i.e. Bybee et al.'s "attention" (no class paper has this exact chart). Blue: voter-voice posts (12,567 issue docs); orange: election headlines (23,045); green: Reuters/Ipsos Sept 17 to 20, registered voters (no bar: no such category). "Foreign policy & Iran war" is mostly Israel aid and AIPAC (9.5 points) and the Iran war (6.9). Issues under 2% and not in the poll are not drawn (education 1.5%). Topic table in the notebook. Descriptive.

Result. Partly: a clear broad picture, but the exact ranking depends on the model run, and it is not the polls'. In the main fit, voter-voice posts give **foreign policy 20.3%** (Israel aid and AIPAC 9.5 points, the Iran war 6.9, Ukraine 2.6, China 0.9, Hormuz 0.5), democracy 11.7%, abortion and social issues 11.5%, immigration 8.8%, taxes 7.3%, health care 7.2%, Trump and his administration 6.6% and ethics 6.2%: the key topics (share ≥ 5%, top 8). The **economy is ninth at 6.2%, 0.06 points behind ethics**, while Pew (29%), Gallup (31%) and Reuters/Ipsos (22%) put it first (rank agreement with the polls, descriptive only: ρ -0.70, -1.0, 0.28); news differs again (economy 14.8%, "Tech and AI" 17.9%, mostly AI and tech-company news; social issues 2.8%).

- **Robust:** the economy gets only 5% to 9% in every version (engagement weights, outlier-reduced assignment, 3 seed refits); foreign policy and democracy are always top three; immigration, taxes, health care and ethics stay in every seed's top 8.
- **Fragile:** the order. Under the outlier-reduced assignment (used by the index) democracy leads (15.3%), then Trump (13.9%) and foreign policy (13.3%). In the seed refits the economy ranks 3rd, 3rd and 4th (8% to 9%) and replaces "abortion and social issues" (3% to 6%), mostly race (609 of 1,446 docs) and religion (348) debates; an index adding the economy is in Q3.

Against Bybee et al.: like their recurrent and emergent topics, in notebook Fig. A's centred 7-day shares foreign policy is recurrent (always ≥ 12.7%) and ethics event-driven (August median 5.2%, 13.3% in Sept 6 to 12; Sept 10 to 20: 115 of 148 posts on PAC money).

Why this result: a poll asks for one problem and "the economy" is the easy answer; people post what is new or contested (a war, voting fights, ICE raids); posters are not a voter sample.

In short. Online political posters talk far less about the economy than polls suggest voters care about (5% to 9% vs 22% to 31%) and more about foreign policy and democracy; the exact order beyond that moves with the model run.

5. Q2: A module that measures sentiment on those issues over time

Readings used: Carvalho and Plastino (2020), Psomakelis et al. (2014) and the autism-awareness slides on why tools disagree and need validation; Souza et al. (2015) for daily sentiment series.

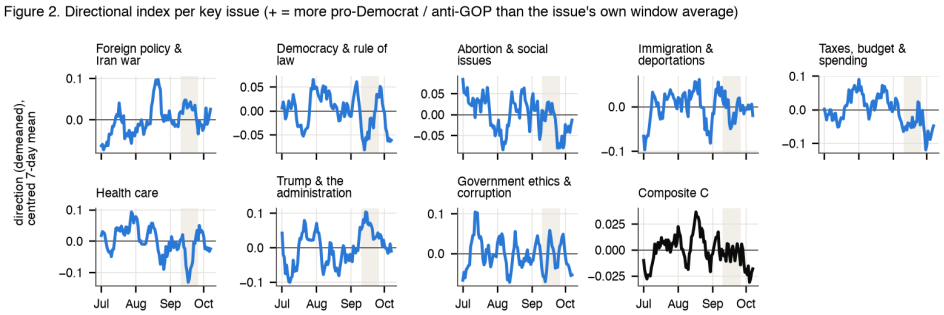


Figure 2. Directional index per key issue and composite C (centred 7-day means; daily series as in Souza et al.): source-weighted mean direction (tweets: NLI; Reddit: RoBERTa x τ), demeaned by issue and source (+ = more pro-Democrat than usual). Grey: Sept 10 to 26. Descriptive.

Result. Partly. Per post, the tools it uses get direction right with fair agreement (macro-F1 0.60, κ 0.38 on 110 social posts, vs 0.22 for a majority guess); the composite uses 18,831 voter docs (median 189 a day; each key issue has 5+ docs on 94% to 100% of days). But **as a daily series it is mostly noise**: over 50 random splits of each day's posts into halves, the half-composites correlate at r = 0.08 on average (range -0.05 to 0.22; Spearman-Brown 0.15). Its versions agree (r 0.94 equal weights, 0.99 expanding weights, 0.99 adding the economy, 0.80 all social posts, 0.75 the 20-doc minimum, 0.71 RoBERTa only) and differ from plain tone (r = -0.22); it hardly trends (first- and second-half means 0.001 and -0.002; sd 0.038).

Against the class readings: as in Carvalho and Plastino (2020) and Psomakelis et al. (2014), tools disagree and the lexicon (VADER) is last for direction and tone.

Why this result: direction is harder than tone (0.60 vs 0.71), and about 189 posts a day are split over 8 issues and 7 sources (median non-empty cell: 3 posts); who posts that day moves the mean more than any shift in mood, and demeaning by source removes only the fixed lean.

In short. The directional module works per post with fair accuracy and a clear rule for which tool is used where, but at a daily horizon it carries little signal (split-half r 0.08), which sets a low ceiling for Q3 and Q4.

6. Q3: Does issue sentiment move with the probability of a Democratic sweep?

Readings used: Souza et al. (2015) for testing both directions and reporting each lag. External: Snowberg, Wolfers and Zitzewitz (2007) on prediction-market prices as probabilities.

Figure 3. Sweep odds and the directional index, in separate panels (no shared axis)

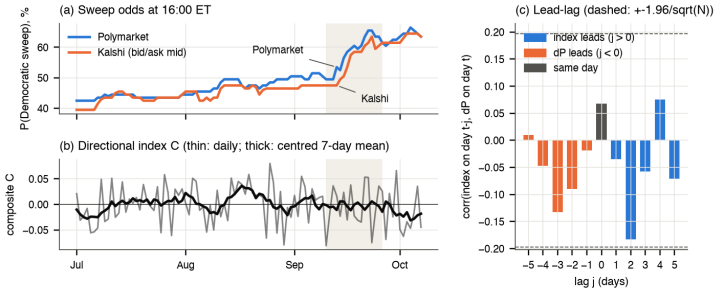


Figure 3. (a) P(Democratic sweep) at 16:00 ET, Polymarket and Kalshi bid/ask mid. (b) Composite C, daily and centred 7-day mean, in its own panel (no second axis). (c) corr(C on day t-j, ΔP on day t), Polymarket, the lead-lag view in Souza et al.; dashed lines are ±1.96/√N. Grey band: the Sept 10 to 26 repricing. Panels (a) and (b) are descriptive; (c) is the lead-lag test, not a prediction.

Result. No. The odds rose from 42.5% (Jul 1) to 63.5% (Oct 7), peak 66.5% on Oct 4, mostly in Sept 13 to 23, but the index does not move with them: **r = 0.067** with the daily change (N = 99). C is stationary (ADF p < 0.001, KPSS p ≥ 0.10), so by the pre-set rule its level is tested against ΔP (the odds have a unit root, ADF p = 0.97).

Table 5. Family A, C_t vs ΔP_t (Polymarket), N = 96 to 99 days

Test	Lag	Pearson r / sum of coefs	HAC p	Permutation p	BH q
Same day	0	r = 0.067	0.58	0.53	0.68
C → ΔP	1 / 2 / 3	-1.07 / -6.84 / -6.79	0.69 / 0.06 / 0.09	0.74 / 0.13 / 0.17	0.69 / 0.31 / 0.31
ΔP → C	1 / 2 / 3	-0.002 / -0.006 / -0.010	0.56 / 0.58 / 0.14	0.63 / 0.59 / 0.30	0.68 / 0.68 / 0.32

- **No test passes the claim rule** (none has q < 0.10); r = 0.067 is far below what 99 days can detect (r* = 0.28). Same-day r is -0.036 on Kalshi, 0.04 with the economy added, and 0.15 (p = 0.16) with the 20-doc minimum; the secondary test in changes (ΔC vs ΔP) is null (r = 0.069, smallest p 0.11).
- The lowest p-values (C → ΔP, lags 2 and 3) have the **opposite** sign; the VAR(3) response agrees at day 2 (-0.20 points, 90% band -0.38 to -0.01), though AIC and BIC both pick 0 lags. It vanishes with the 20-doc minimum (p = 0.95). In the grid 21 of 93 tests have p < 0.10, all negative, but shifted index series give 17.0 such hits on average (21+ in 30%), so the count is chance; the strongest (trading-day ΔP → C, lag 3, within-spec q = 0.016) is wrong-signed too.
- The ceiling is low anyway: the two venues' daily changes correlate at only 0.40, and the odds do not move on 57% of days.

Against Souza et al.: they find Twitter sentiment Granger-causes some retail stocks' returns at a 1-day lag; here neither direction works at lag 1, and the lag-2 pattern has the wrong sign.

Why this result: the repricing coincided with polls (NYT/Siena Sept 15, state Senate polls), a ratings change (Cook, Sept 23) and high gas prices ($4.48 on Sept 24), which traders read directly, while posters argue about the same war and courts daily; a noisy index against odds that move in steps on few days leaves little room for a link.

In short. Text sentiment on the key issues did not lead, follow or move with the sweep odds at a daily horizon (r = 0.067); the one pattern is fragile and wrong-signed, and the test had little power to begin with.

7. Q4: Which stocks are exposed, and does the basket move with sentiment?

Readings used: Rigobon and Sack (2003), the class reading on how a political risk moves asset prices, with Rigobon (2003) for the identification idea. External: MacKinlay (1997), Knight (2006), Snowberg et al. (2007).

A **basket** is a group of stocks held in equal weights: a **long leg** expected to gain from a sweep, a **short leg** expected to lose, and **long minus short**, which nets out the market and leaves the election bet; returns are **market-adjusted**, AR = r - β x SPY (β from Aug 2025 to Jun 2026).

Table 6. The mid-term basket (signs fixed before test-window data; γ = % AR per point of P(sweep), Aug 2025 to Jun 2026, N = 229)

Leg	Group: names	Why a sweep matters, with a Republican veto	γ	Expected sign (pre-window)
Long	ACA/Medicaid insurers: OSCR, CNC, MOH	ACA credits expired Dec 31, 2025; a House extension died in the Senate	+0.36	3 of 3
Long	Hospitals: HCA, THC, UHS, CYH	exchange-loss hits (HCA $1.0 to 1.2B); HCA, UHS, CYH cut 2026 guidance	+0.14	4 of 4
Long	Guns: RGR, SWBI	+12% and +18% when the 2021 Georgia runoffs gave D the Senate	+0.23	2 of 2
Short	Detention: GEO, CXW	ICE was 47.6% and 35% of 2025 revenue; money needs Congress	+0.07	0 of 2
Short	Defense: LMT, NOC, GD, HII	$350B of the $1.5T FY27 request needs a GOP reconciliation bill	-0.01	1 of 4
Short	Crypto: COIN, HOOD	CLARITY Act failed cloture on Sept 15, no Democratic votes	+0.14	0 of 2

The long leg has some pre-window support (γ = +0.24, SE 0.12, p = 0.055; CNC and MOH clear of zero), gone on the Jan to Jun venue-average window (γ 0.03, p = 0.82); the **short leg rests on the policy mechanism only** (1 of 8 names with the expected sign; γ +0.05, p =

0.53). Oil, clean energy and tariff retail (10 names) are a "weak" sub-basket, as their lever is fixed by statute or executive action; pharma and banks are out (both parties push drug prices; regulators set capital rules). Per-name γ in the notebook.

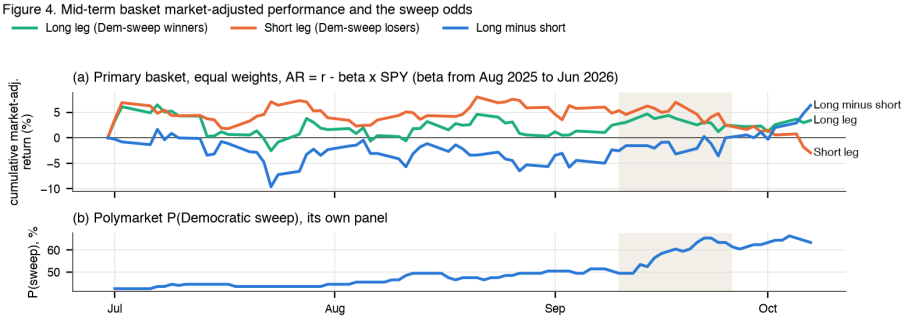


Figure 4. (a) Cumulative market-adjusted return of the equal-weighted long leg, short leg and long minus short, from 0 on Jun 30. (b) Polymarket P(sweep) in its own panel. Grey: the Sept 10 to 26 repricing. No class paper has this exact figure. Descriptive; tests in the text.

Table 7. Sentiment index C vs the basket's market-adjusted long-minus-short return (trading days)

Test	Lag	Pearson r / sum of coefs	HAC p	Permutation p	BH q
Same day	0	r = -0.015 (Spearman 0.002)	0.49	0.89	0.57
C → basket	1 / 2 / 3	-6.05 / -11.10 / -8.72	0.21 / 0.18 / 0.23	0.23 / 0.32 / 0.48	0.40 / 0.40 / 0.40
Basket → C	1 / 2 / 3	-0.002 / -0.003 / 0.003	0.38 / 0.67 / 0.21	0.48 / 0.70 / 0.54	0.53 / 0.67 / 0.40

Result. No, the sentiment index and the basket are not related: **r = -0.015** (N = 69) and no test passes; the 8 of 49 robustness hits are fewer than a shifted null gives (10.4), and the three within-spec hits (the factor-adjusted weak basket leading sentiment at lags 1 to 3, each q = 0.073) fail the permutation (p 0.13 to 0.39) and episode checks. Nor does the basket move with the odds: slope on ΔP 0.05 (SE 0.16, r = 0.03) on Polymarket, 0.17 (p = 0.22) on Kalshi; the Rigobon and Sack estimate from the 12 repricing days (ΔP variance 5.0 times higher) is +0.34, 95% CI -0.61 to 1.24 (0.31 without the Fed-hike day; days picked from the odds, not dated news).

- **The window total is an endpoint effect.** Long minus short ended at **+6.4%** (long +3.4%, short -2.9%; +4.3% against sector factors), but it was -2.4% on Sept 9, gained 2.3% in the repricing (odds 50.5% on Sept 9 to 63.5% on Sept 25), stood at **-0.3% on Oct 1** after the whole 21-point rise, then gained **6.6% in Oct 2 to 7** (1.6 on Oct 7) while the odds stayed within 63.5% to 66.5% and ended at 63.5%. So the total says nothing about the election.
- In the window only 8 of 17 names have ΔP slopes of the expected sign; per name (BH over 27) only RUN (+0.82, expected) and DLTR (-0.62, opposite) track ΔP (both q = 0.002, weak sub-basket); insurers and hospitals do not (slopes -0.22 to +0.30). OSCR's sign is unclear (it gained ACA members); without it the basket made +5.8%. Single names and sectors dominate (THC +30%, CYH -26%, MOH -15%; the weak sub-basket -41% market-adjusted but -8% against its factors, as oil rose (USO +35%) and clean energy fell (ICLN -15.5%)).

Against Rigobon and Sack: they sign war risk's effect (equities down, oil up) from high-variance news days; ours also carry 5 times the ΔP variance, but the basket's response is too noisy to sign. *Against Knight (2006):* he finds firm-level effects in 2000 (tobacco +13%, alternative energy -16%); our pre-window long leg fits that, the test window does not.

Why this result: with a Republican veto a sweep changes few laws, so the effect per point is small and single-stock noise hides it; a mostly-noise daily index caps any correlation.

In short. I can name exposed stocks with a policy reason, and the long side (insurers, hospitals) had some pre-window price support, but in the window neither the sentiment index (r = -0.02) nor the odds (r = 0.03) track the basket, and its +6.4% total is an endpoint effect.

8. Q5: What I conclude

Readings used: all of the above; the numbers are from Sections 4 to 7.

1. **What online posters argue about is not the polls' agenda.** In 12,567 voter-voice issue posts the economy is 5% to 9% in every model version against 22% to 31% in every poll, and foreign policy and democracy are always top three. Robust; the finer order is fragile. For a PM, polls and social text answer different questions; text shows what is argued about this week.
2. **Media cover a different agenda again.** In 23,045 election headlines the economy is 14.8% and markets or tech 17.9%, but social issues only 2.8% (posters 11.5%, a fragile share). Robust across outlet groups: every group is further from posters than posters' own shared links are (E1).
3. **A directional political-sentiment index can be built with fair per-post accuracy (macro-F1 0.60 vs 0.22 for a guess), but daily it is mostly noise** (split-half r 0.08, 189 docs a day). Do not trade on a daily political mood index of this size.
4. **No daily link between sentiment and the sweep odds** (r = 0.067, N = 99; 0 of 7 pre-registered tests pass). The low p-values have the wrong sign, vanish under the pre-registered 20-doc minimum, and their count is what chance gives (P = 0.30). Correlation and Granger are not causality in any case.
5. **No link between sentiment and the mid-term basket** (r = -0.02, N = 69; 0 of 7 tests pass), nor between the basket and the odds (r = 0.03). Its +6.4% is an endpoint effect (-0.3% on Oct 1), the short leg has no price support, and the long leg's is fragile. With a Republican veto the stakes per point of sweep probability are small; a PM should not read this basket as an election hedge.

In short. What people talk about is measurable and different from what polls and media show, but a daily political mood index of this size moves with neither the sweep odds nor the exposed stocks, and the basket's window gain is an endpoint effect, not an election effect.

9. Extra questions that came up

Readings used: Sacerdote et al. (2020) for E1; Souza et al. (2015) for E2's per-issue tests; Rigobon and Sack (2003) for E3's event window.

E1. Do media and voters care about the same issues? *Result:* no, business media least. Agenda distance from voter-voice posts (TVD; 0 = same, 1 = disjoint): posters' shared links 0.16, all news 0.27 (incl. the US feed), US left 0.21, right 0.21, centre 0.28, international 0.29, business 0.51. Right-leaning outlets rank issues closest to posters (Spearman 0.75; left 0.65, business 0.50), and the economy's share rises

from posters (6.2%) to right (8.2%), left (9.9%), centre (16.0%) and business outlets (26.9%). *Why this result:* likely, partisan outlets cover the fights partisans post about, while business media cover prices, rates and markets; as in Sacerdote et al., the outlet group shapes what you see. *In short:* the further an outlet is from partisan politics, the further its agenda is from posters'.

E2. Which single issue tracks the sweep odds most? *Result:* none; over 9 issues (key 8 plus the economy) x 3 tests (BH over 27), min q = 0.57 (Trump same-day p = 0.14; economy p = 0.27). *Why this result:* each issue's index has even fewer docs. *In short:* no hidden link by issue.

E3. What happened during the Sept 10 to 26 repricing? *Result:* the odds rose 11 points in 17 days; attention shifted (vs Aug 24 to Sept 9): up for foreign policy (+4.0 pts), economy (+3.5), social issues (+3.1), ethics (+2.7); down for democracy (-4.9), tariffs (-4.3), health care (-3.6). The index's mean did not move (0.002 before, -0.003 during). *Why this result:* the repricing came with polls and a ratings change, which traders watch; attention moved, each side kept its view. *In short:* the repricing shows in what people talk about, not which side they favour.

10. Based on latest in-class discussions

Readings used: the Oct 3 lecture "Topic modeling: the 2026 midterms and equity markets" (scenario values $\Delta V \approx \Sigma_s \Delta Pr(s)\, V(s)$, slide 24; exposure-weighted attention after Hassan et al., slides 22 to 23; falsification, slide 28) and the meme-stock deck (attention is not sentiment; does trading confirm the story; after Alvarez et al., ICWSM 2022). External: Karpoff (1987). The tests were written down first (`config.PREREG_CLASS`, Oct 10, 19:27 ET) but after the main results, so they are exploratory and change nothing above; † marks numbers added after an independent review, not in that record.

- **How.** (1) The five Polymarket outcomes are mutually exclusive (16:00 prices sum to 1.00 on average, 0.96 to 1.03), so P(D House) = P(D sweep) + P(R Senate, D House) and P(D Senate) = P(D sweep) + P(D Senate, R House) (Kalshi: DD + DR, DD + RD). (2) Each strong group vs its own issue (health care, immigration, foreign policy, tech and AI, guns): the issue's index (Q2) and attention (its share of voter issue docs), signed so + is expected. (3) Attention $A_t = \tfrac{1}{2}[z(\log \text{Reddit posts}) + z(\log \text{tweet rate})]$; a Twitter page of n tweets over T minutes gives a rate (n - 1)/T, one over the mean gap (pages hold 20 tweets, so counts say nothing; Reddit posts were collected in full). (4) Kalshi sweep volume per 16:00 day. (5) 10,000 random 9/8 sign splits of the 17 names, and an Indian-politics text index.

Table 8. Five extensions (claim rule as in Table 5; basket slopes in % per point; every table in the notebook)

Extension	Result
(1) Scenario odds	Mostly a Senate repricing: Senate 44% → 64%; House 82% → 90% (range 82% to 94%), about a quarter of the sweep's rise in log terms (40% over Sept 10 to 26)†. Basket on both jointly: House +0.13 (p = 0.19), Senate +0.11 (p = 0.48) pre-window; +0.08 (p = 0.61), -0.01 (p = 0.97) in the window; pre-window House alone† +0.21 (p = 0.08; 0.11 without the 25 days around 19 stale prices). C vs House: same-day r = 0.19 (q = 0.05) passes four checks but not the episode drop (p = 0.23), and is -0.11 on Kalshi; lagged hits wrong-signed. No claim
(2) Issue-matched	7 of 35 direction tests p < 0.10 (3 wrong-signed†), none after pooled BH; closest: more pro-Democrat foreign-policy sentiment leads defense stocks down (lag 1, p = 0.004, permutation 0.04, q = 0.14). Attention vs size of group moves: none (min q 0.39). Mismatched pairs correlate nearly as much (descriptive: mean abs r 0.08 vs 0.11; the largest, -0.28, is mismatched)
(3) Attention	Up in the repricing† (mean +0.35 vs -0.09) but no daily link: r = 0.03 (Kalshi 0.23, p = 0.16), 0 of 7 p < 0.10, also without Oct 6 to 7 and in changes† (ADF p = 0.06). Parts pull apart†: tweet rate r = 0.20 (HAC p < 0.001, shift 0.17), Reddit posts -0.15 (shift 0.38)
(4) Volume	Odds moves are traded: log volume vs abs ΔP r = 0.50 on Kalshi, 0.44 on Polymarket (HAC p < 0.001; permutation 0.012, the floor, on both†), 0.42 and 0.34 without Sept 10 to 26†; partly mechanical on Kalshi, where trades move the mid. Volume 4.7x higher Sept 10 to 26; open interest 1.0M → 2.7M contracts. A_t vs volume r = 0.25 (q = 0.49): no claim
(5) Placebos	Pre-window (the period used to approve the basket) the real signs beat 99.8% of name splits (γ 0.18, itself p = 0.21†) but only 92% of whole-group sign flips† (5 of 62 beat it); in the window they sit at the 65th (ΔP) and 42nd (C) percentiles. The placebo text index has 2 of 7 p < 0.10 (min q 0.11), as many as the real index, whose two are wrong-signed

Why these results: the House was 82% to 94% Democratic all window, and P(Democrats win at least one chamber)†, with the prices normalised to sum to 1, rose only from 83% to 91% (peak 93%). Blocking the GOP agenda (reconciliation, the CLARITY Act, ICE money) and oversight need one chamber (a bill must pass both, U.S. Constitution Art. I, Sec. 7; nominations and treaties need the Senate), so those levers were mostly priced before July, and most of the sweep's rise came from the Senate. Volume rising with the size of odds moves is consistent with Karpoff (1987): the moves were traded, which does not make them informed. In 69 to 99 days only r of about 0.28 to 0.33 is detectable (Section 11), and across 20+ comparisons a few of 0.2 to 0.3 arise by chance, as the mismatched pairs and the placebo index suggest.

In short. The class ideas sharpen the null rather than overturn it: most of the odds' rise was the Senate's, while the basket's blocking levers need only the already-likely House; odds moves are traded, but our attention measure passes no test; nothing issue-matched survives pooled BH; and a placebo index gives as many near-misses as ours. In-sample, before July, the policy signs sat at the 92nd percentile of group flips (mainly the long leg); in the window they carried no information.

11. Limitations

- **Sample and measurement.** Twitter is a paid sample of engaged tweets (7 empty windows); Reddit is six political subs, some heavily moderated; news is capped and dated only; none of it samples registered voters. 40% of docs are topic outliers; the taxonomy and all labels come from Claude subagents, not humans (κ 0.96 between two subagents, which may share biases); the index needed a disclosed deviation (5 docs per issue-day, not 20); τ counts "conservatives" and "liberals" as party names; tools were chosen and scored on the same 150 items; the topic model saw the whole window's text, never prices.
- **Power, timing and causality.** 99 days (69 trading days) detect only r ≥ 0.28 (0.33); Granger and correlation are not causality; confounders include the Fed hike, the Iran war's oil effect and earnings. Kalshi's snapshot is about an hour old on 2 window days, Polymarket's 1 to 6 hours on 14 pre-window days (without the 23 affected days the long-leg γ is 0.234, p = 0.064). No election-day test (due before Nov 3).

References. Class readings: Bybee, Kelly, Manela and Xiu (2024), JF · Rigobon and Sack (2003), NBER WP 9609 · Rigobon (2003), REStat · Souza, Kolchyna, Treleaven and Aste (2015) · Sacerdote, Sehgal and Cook (2020), NBER WP 28110 · Carvalho and Plastino (2020) · Psomakelis et al. (2014) · autism-awareness slides; Bagheri; Yener (2020) · lecture "Topic modeling: the 2026 midterms and equity markets" (Oct 3, 2026) and the meme-stock deck, with Hassan et al. and Alvarez et al. (ICWSM 2022) as cited there. External: Karpoff (1987), JFQA · Knight (2006), JPubE · Snowberg, Wolfers and Zitzewitz (2007), QJE · MacKinlay (1997), JEL · Grootendorst (2022) · Blei, Ng and Jordan (2003) · Röder et al. (2015) · Laurer et al. (2023) · Loureiro et al. (2022) · Hutto and Gilbert (2014) · Cohen (1960) · Spearman (1910); Brown (1910) · Granger (1969) · Newey and West (1987) · Benjamini and Hochberg (1995) · Said and Dickey (1984); Kwiatkowski et al. (1992) · Hubert and Arabie (1985) · Hulley et al. (2013) · Levin, Peres and Wilmer (2009) · Yuan and Shou (2024). Data and facts: Polymarket, Kalshi (APIs, rules); twitterapi.io; Arctic Shift; Google News RSS; Pew (Jul 23, 2026); Gallup MIP (Sept 2026); Reuters/Ipsos (Aug 28 to 31, Sept 17 to 20); KFF (Oct 6); 10-K/10-Q filings (GEO, CXW, CNC); Defense News; Breaking Defense; Newsweek; Fortune; NPR; AP; Federal Reserve (Sept 16); DOI; IRS.